

Digital Research Toolkit for Linguists

Week 7: Data visualization and exporting

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Psycholinguistics and Cognitive Modeling Lab

Homework

Name your files with your name and assignment number.

Use  | >  (at all and instead of %>%).

→ if you're going to cheat, be smart, read the slides & code

You don't need to copy the results, I can just run your code.

Good variable naming!

Course credit and exam

6 credit points

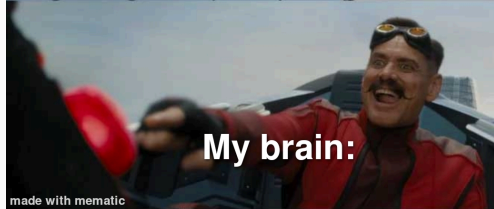
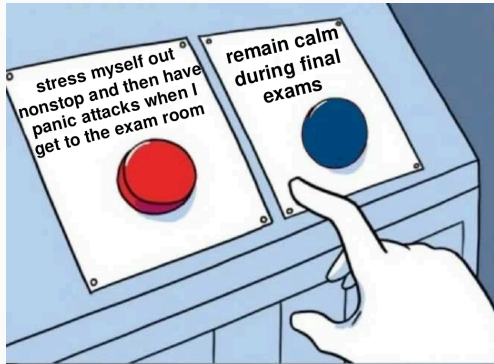
- ✓ written exam (graded, multiple choice-ish)
 - ✓ complete $n - 2$ home-works (ungraded)
 - ✓ poster (ungraded).
- ✗ complete $< n - 2$ home-works.

9 credit points

- ✓ written exam (graded, multiple choice-ish)
 - ✓ complete $n - 2$ home-works
 - ✓ poster
 - ✓ poster presentation.
- ✗ complete $< n - 2$ home-works.

Register (or deregister) in Campus!

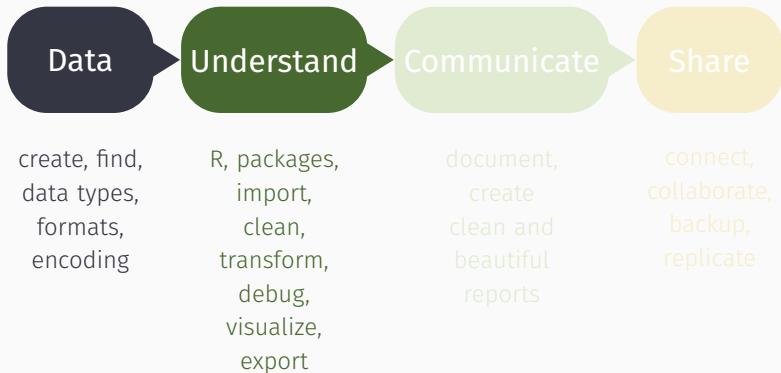
Exam date: Tuesday, July 22nd at 11:30



Questions?

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Data visualization

Data visualization

is an interdisciplinary field. It's the graphical representation of data & information in order efficiently communicate complex relationships & insights in a comprehensible way.

Goals

🗨️ Communication & understanding

⚙️ Analysis & exploration

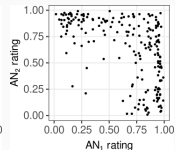
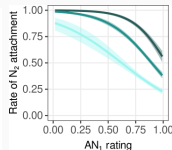
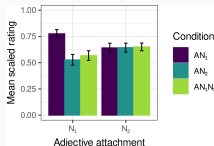
⚖️ Decision making



Visualization types

boil down to tables, bars, lines, and points, and a mix of all four.

item	sentence	N1	N2	unclear
3	sommerliche Arbeitskleidung	p	q	p
11	silberne Bahnhofsuhr	p	q	p
12	amtliche Baugenehmigung	p	q	p
16	hohler Bienenstock	p	q	p
19	gefüttertes Brillenetui	p	q	p
22	lackiertes Bücherregal	p	q	p
23	evangelischer Büroangestellter	p	q	p



The 4 Principles of Accessibility

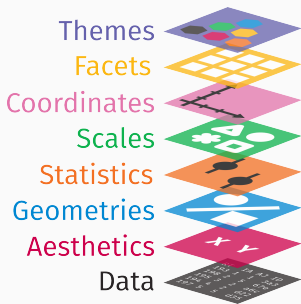
Web Content Accessibility Guidelines

- P Perceivable
- O Operable
- U Understandable
- R Robust

Plotting in R

ggplot2

a layered grammar of graphics

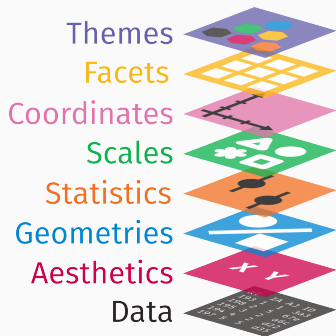


esquisse

GUI for exploring data based on ggplot2



ggplot2: A Layered Grammar of Graphics



Created by Hadley Wickham.

Design and construct graphics in a structured manner from data upwards.

“A **layer** is a collection of geometric elements and statistical transformations” (Wickham et al. n.d.).

Assembly

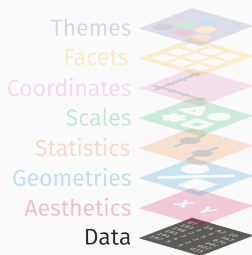


1. Create a plot object
2. Add aesthetic mapping
3. Add geometric objects
4. Add appearance and statistics
- ... repeat 3 and 4 until you're happy
5. Print and profit!

```
my.plot <- ggplot()+  
  aes()+  
  geom_*()+  
  coord_*() + theme()
```


Data

Typically one row per observation for plotting.



	CONDITION	IA	MeanRT	SD
	<i><chr></i>	<i><chr></i>	<i><dbl></i>	<i><dbl></i>
1	implausible	1	823.	119.
2	implausible	2	759.	107.
3	implausible	3	818.	179.
4	implausible	4	<u>1023.</u>	133.
5	plausible	1	781.	106.
6	plausible	2	700.	64.8
7	plausible	3	783.	115.
8	plausible	4	<u>1092.</u>	137.

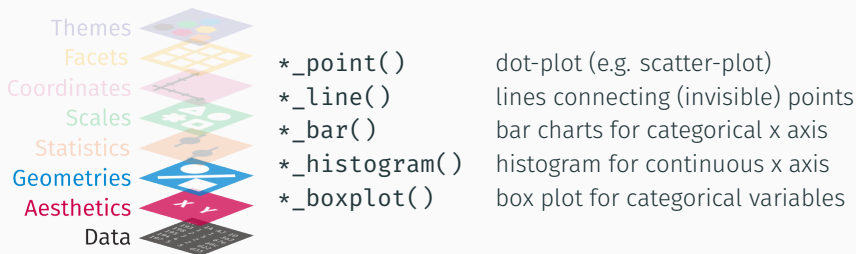
Aesthetics

Aesthetics `aes()` translates data into visual elements:



Geometries

Geometries (`geom_*()`) provide the shapes and patterns to the data:





Statistics (`stat_*()`) compute and add statistical transformations to the data (e.g. means, counts, linear models). We won't use it in this course.

Scales

Scales (`scale_*()`) turn your data into visual elements and add tools like axes and legends.



You can prettify your plots with them (e.g. legend position, axis limits, custom fill and color).

```
scale_y_continuous()  
scale_colour_discrete()  
scale_shape_manual()  
scale_fill_gradient()  
scale_x_sqrt()
```

Coordinates

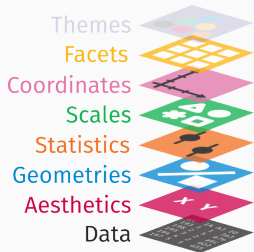
Coordinates (`coord_*()`) change the coordinate system and axes.



<code>*_cartesian()</code>	sets the limits of the coordinate system
<code>*_polar()</code>	circular plots
<code>*_map()</code>	map projections
<code>*_flip()</code>	flips the coordinates

Facets

Facets (`facet_*`) divide your plot into smaller sections across panels.

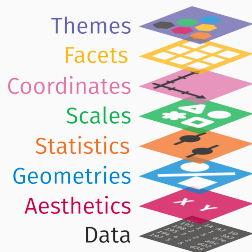


`*_wrap()` based on one variable, it arranges plots in a single row or column, adding more rows/columns as needed

`*_grid()` does the same, but based on two variables

Themes

Themes (`theme_*()`) fine-tune the overall appearance and style:

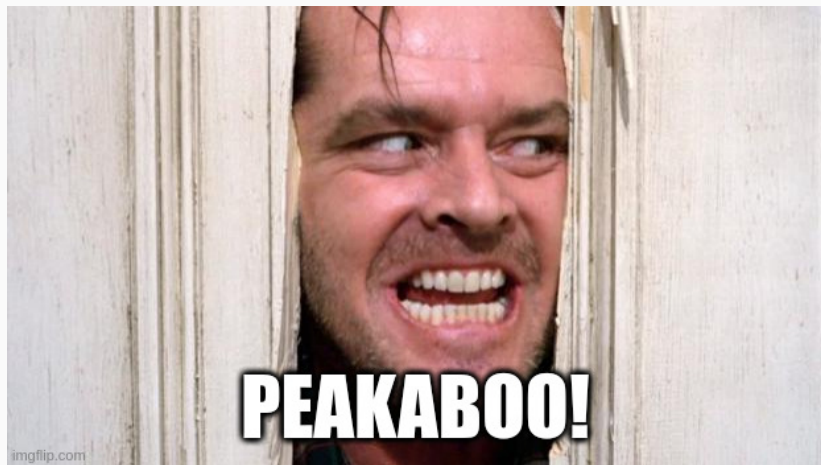


- fonts,
 - background color,
 - line width,
 - legend,
 - axis and tick label sizes and colors,
- ... and so much more!

Honorable mention: Labels

`labs()` allows you to set titles for the x-axis, y-axis, plot title, and other labels.

<code>x</code>	x-axis
<code>y</code>	y-axis
<code>title</code>	plot title
<code>subtitle</code>	plot subtitle
<code>caption</code>	an additional caption
<code>colour</code>	color legend title
<code>size</code>	size legend title
<code>fill</code>	fill legend title



esquisse: a rough or preliminary sketch



esquisse lets you explore data interactively. Uses **ggplot2** for visualization.

You can export the generated graph and save the code to generate it.

Available in many languages (incl. French, Turkish, Korean, German).

It has its limitations but is useful to get a first impression/overview.

Questions?

Wrap-up

- ✓ Data visualization with `ggplot2`
- ✓ Building layers of graphics
- ✓ Previewing plots with `esquisse`
- ▶▶ Using color, lying with plots

- ❓ Complete assignment 7 (→ ILIAS)
- ▶▶ Plots: Best practices, lying, exporting.

References



Wickham, Hadley, Danielle Navarro, and Thomas Lin Pedersen (n.d.). *ggplot2: Elegant Graphics for Data Analysis*. URL: <https://ggplot2-book.org/>.